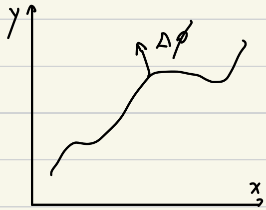
2- order PDE

$$a(x,y)\,u_{xx} + b(x,y)\,u_{xy} + c(x,y)\,u_{yy} = f(x,y,u,u_x,u_y)$$

Discriminant $\Delta = b^2 - 4ac$

Characteristic line.

$$T:\ \phi(x,y)=0$$
$$\nabla\phi = (\phi_x, \phi_y)$$



fix $(x,y) = (x_0, y_0)$

$\eta = \phi(x,y) \qquad \xi = \psi(x,y) \qquad$ where $\begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} \neq 0$

$u_x = u_\xi \xi_x + u_\eta \cdot \eta_x \qquad u_y = u_\xi \xi_y + u_\eta \eta_y$

$$u_{xx} = u_{\xi x}\xi_x + \cdots + u_\eta \eta_{xx}$$
$$= \xi_x^2 u_{\xi\xi} + \cdots + \eta_{xx} u_\eta$$

coefficient of $u_{\eta\eta}$:

$$A\eta_x^2 + B\eta_x\eta_y + C\eta_y^2$$

$$u_{xy} = u_{\xi y}\xi_x + \cdots + u_\eta \eta_{xy}$$
$$= \xi_x\xi_y u_{\xi\xi} + \cdots + \eta_{xy} u_\eta$$

coefficient of $u_{\xi\xi}$:

$$A\xi_x^2 + B\xi_x\xi_y + C\xi_y^2$$

$$u_{yy} = u_{\xi y}\xi_y + \cdots + u_\eta \eta_{yy}$$
$$= \xi_y^2 u_{\xi\xi} + \cdots + \eta_{yy} u_\eta$$

法向二阶导?

if $A\eta_x^2 + B\eta_x\eta_y + C\eta_y^2 = 0$. write $\phi(x,y) = y - f(x)$

$A f'^2 - B f' + C = 0$

$\Delta = B^2 - 4AC$

$\Delta < 0$ Elliptic 2 I 2 B

$\Delta = 0$ parabolic 1 I 2B

方程在曲线法向 (ϕ_x, ϕ_y) 上退化

$\Delta > 0$ hyperbolic 2 B

$x^2 y'' + axy' + by = 0$ $t = \ln|x|$ $y(x) = u(t)$

$y' = \frac{dy}{dx} = \frac{du}{dt}\frac{dt}{dx} = \dot{u}\cdot\frac{1}{x}$

$y'' = \frac{dy'}{dx} = \frac{d\dot{u}\cdot\frac{1}{x}}{dx} = \frac{\frac{1}{x}d\dot{u} + \dot{u}\cdot(-\frac{1}{x^2})dx}{dx} = \frac{1}{x^2}\ddot{u} + (-\frac{1}{x^2})\dot{u}$

$u'' + (a-1)u' + bu = 0$ 欧拉方程

Heat equation.

$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ Initial condition $u(x,0) = f(x)$

BCs $u(0/L, t) = 0$

$u(x,t) = X(x)\,T(t)$

$$X(x)T'(t) = X''(x)T(t)$$

$$\frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} = \lambda$$

Workshop 2.

1. $(1+x)y'' - y = 0$

$$y = \sum_{n=0}^{\infty} a_n x^n$$

$$y'' = \sum_{n=0}^{\infty} n(n-1) a_n x^{n-2}$$

$$= \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n$$

$$(1+x) \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n - \sum_{n=0}^{\infty} a_n x^n$$

$$= \sum_{n=0}^{\infty} \left[(n+2)(n+1) a_{n+2} + (n+1)n a_{n+1} - a_n \right] x^n = 0$$

$$(n+2)(n+1) a_{n+2} + (n+1) n a_{n+1} = a_n$$

accept $[0,-1)$

2. $2x^2 y'' + 3xy' - (x+1)y = 0$

$$y'' + \frac{3}{2x} y' - \frac{(x+1)}{2x^2} y = 0$$

$$\lim_{x \to 0} \frac{3}{2x} \cdot x = \frac{3}{2}, \quad \lim_{x \to 0} -\frac{x+1}{2x^2} \cdot x^2 = -\frac{1}{2}$$

$$r(r-1) + \frac{3}{2} r - \frac{1}{2} = 0$$

$$2r^2 + r - 1 = 0$$

$$y = \sum_{n=0} a_n x^{n+p} \qquad y' = \sum_{n=0}^{\infty} a_{n+1} (n+p) x^{n+p}$$

$$y'' = \sum_{n=0}^{\infty} a_{n+2} (n+p)(n+p-1) x^{n+p}$$

$$\sum_{n=0}^{\infty} \left[2 a_n (n+p-2)(n+p-3) + 3 a_n (n+p-1) - a_n - a_{n-1} \right] x^{n+p} = 0$$

$$a_n \left[2(n+p-2)(n+p-3) + 3(n+p-1) - 1 \right] = a_{n-1}$$

$$= a_n \left[2(n+p)^2 - 2(n+p) + 2 \right]$$

Workshop 2.

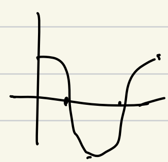
(a) $$f(x) = \begin{cases} -1 & x \in (-\pi, -\frac{\pi}{2}) \\ 1 & x \in (-\frac{\pi}{2}, \pi) \end{cases}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos\left(\frac{n\pi x}{\pi}\right) dx \qquad n = 0, 1, 2 \cdots$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin\left(\frac{n\pi x}{\pi}\right) dx \qquad n = 1, 2, \cdots$$

$$a_n = \frac{1}{\pi}\left(\int_{-\pi}^{-\frac{\pi}{2}} -1 \cos nx \, dx + \int_{-\frac{\pi}{2}}^{\pi} \cos nx \, dx \right)$$

$$= \frac{1}{\pi} \int_{-\frac{\pi}{2}}^{\pi} \cos nx \, dx = \frac{2}{\pi n} \sin \frac{n\pi}{2}$$



1 -2 1 0

$$a_0 = 1$$

$$b_n = \frac{1}{\pi}\left(\int_{-\pi}^{-\frac{\pi}{2}} -\sin nx \, dx + \int_{-\frac{\pi}{2}}^{\pi} \sin nx \, dx \right)$$

$$= \frac{1}{\pi} \left(\frac{1}{n} \cos nx \Big|_{-\pi}^{-\frac{\pi}{2}} - \frac{1}{n} \cos nx \Big|_{-\frac{\pi}{2}}^{\pi} \right) = \frac{2}{\pi n} \left(\cos \frac{n\pi}{2} - \cos n\pi \right) = \begin{cases} 1 & n = 1, 3, 5 \cdots \\ -1 & n = 2, 6, \cdots \\ 0 & n = 4, 8, \cdots \end{cases}$$

Problem Sheet I.

4.

$$W(x) = W(x_0)\exp\left(-\int_{x_0}^{x} a_{n-1}(t)\,dt\right)$$

$$W'(x) = -p(x)\,W(x)$$

8. for $y'' + p(x)y' + q(x)y = g(x)$

y_1 is a solution of the Homogeneous ODE

the general solution $y = v(x)\,y_1(x)$ satisfies

$$v'' + \left[\frac{2y_1'(x)}{y_1(x)} + p(x)\right]v' = \frac{g(x)}{y_1(x)}$$

9. $$r(x)\left(v'' + \left[\frac{2y_1'(x)}{y_1(x)} + p(x)\right]v'\right) = \left[r(x)\,v'(x)\right]'$$

where $r(x) = \exp\left(\int p(x) + \frac{2y_1'(x)}{y_1(x)}\,dx\right) = [y_1(x)]^2\exp\left(\int p(x)\,dx\right)$

11. $y'' + p(x)y' + q(x)y = g(x)$ $\quad y(x_0) = 0 \quad y'(x_0) = 0$

$$y = \int_{x_0}^{x}\left(\frac{y_1(s)\,y_2(x) - y_1(x)\,y_2(s)}{W(y_1, y_2)(s)}\right)g(s)\,ds$$

especially when $p(x) = -a-b$ $\quad q(x) = ab$

$$y = \int_{x_0}^{x}\frac{e^{b(x-s)} - e^{a(x-s)}}{b-a}\,g(s)\,ds = \int_{x_0}^{x}K(x-s)\,g(s)\,ds.$$

12. the choice of $u_1' y_1 + u_2' y_2 = 0$ is without loss of generality.

Problem Sheet 2.

7. (b) $3xy'' + 2y' + xy = 0$

$$y'' + \frac{2}{3x} y' + \frac{1}{3} y' = 0$$

$$\lim_{x \to 0} \frac{2}{3x} \cdot x = \frac{2}{3} \qquad \lim_{x \to 0} \frac{1}{3} \cdot x^2 = 0$$

$$r(r-1) + \frac{2}{3} r = 0$$

$$r = 0 \ \& \ \frac{1}{3}$$

$$y'' = \sum_{n=0}^{\infty} (n+r)(n+r-1) a_n x^{n+r-2}$$

$$= \sum_{n=-2}^{\infty} (n+r+2)(n+r+1) a_{n+2} x^{n+r}$$

$$y' = \sum_{n=-1}^{\infty} (n+r+1) a_{n+1} x^{n+r}$$

$$\sum_{n=0}^{\infty} x^{n+r} \Big(3(n+r+1)(n+r) a_{n+1} + 2(n+r+1) a_{n+1} + a_{n-1} \Big)$$

$$+ 3r(r-1) a_0 x^{r-1} + 3(r+1) r a_1 x^r + 2r a_0 x^{r-1} = 0$$

10 Parseval Identity

$$[f(x)]^2 = \frac{a_0^2}{4} + a_0 \sum a_n \cos + b_n \sin + \sum\sum a_n a_m \cos\cos + b_n b_m \sin\sin$$

$$+ \sum\sum 2 a_n b_m \cos \sin$$

$$\frac{1}{L} \int_{-L}^{L} [f(x)]^2 = \frac{a_0^2}{2} + \sum\sum a_n a_m \delta_{nm} + b_n b_m \delta_{nm} = \frac{a_0^2}{2} + \sum (a_n^2 + b_n^2)$$

PS 4.

12 $\nabla^2\psi = \psi \qquad \psi(0,y)=0 \qquad \left.\frac{\partial\psi}{\partial y}\right|_{y=0}=0 \qquad \left.\frac{\partial\psi}{\partial y}\right|_{y=\pi}=0 \qquad \psi(\pi,y)=1-\frac{y}{\pi}$

$$\psi = X(x)\,Y(y)$$

$$X''(x)\,Y(y) + X(x)\,Y''(y) = X(x)Y(y)$$

$$\frac{Y''(y)}{Y(y)} = \frac{X-X''}{X} = -\lambda$$

$$Y'(0) = Y'(\pi) = 0$$

$$\lambda = n^2 \quad , \quad n=0,1,2\cdots$$

$$X'' = (1+n^2)X$$

$$X(x) = B\sinh\left(\sqrt{1+n^2}\,x\right)$$

$$\psi(x,y) = A_0\sinh(x) + \sum_{n=1}^{\infty} A_n \sinh\left(\sqrt{1+n^2}\,x\right)\cos(ny)$$

$$\psi(\pi,y) = A_0\sinh(\pi) + \sum_{n=1}^{\infty} A_n \sinh\left(\sqrt{1+n^2}\,\pi\right)\cos(ny) = 1-\frac{y}{\pi}$$

$$A_n \sinh\sqrt{1+n^2}\,\pi = \frac{2}{\pi}\int_0^{\pi}\left(1-\frac{y}{\pi}\right)\cos(ny)\,dy$$

$$= \frac{2\left[1-(-1)^n\right]}{n^2\pi^2}$$

$$\psi(x,y) = \frac{\sinh(x)}{2\sinh(\pi)} + \frac{2}{\pi^2}\sum_{n=1}^{\infty}\frac{\sinh\left(\sqrt{1+n^2}\,x\right)\cos(ny)\left[1-(-1)^n\right]}{n^2\sinh\left(\sqrt{1+n^2}\,\pi\right)}$$

PS 5

6. Solve $\nabla^2 u(x,y) = 1$ $\quad u(1,y) = u(x,1) = u(0,y) = u(x,0) = 0$

(a) first consider $\nabla^2 v(x,y) = 0$ $\quad v(0,y) = v(1,y) = 0$

$$u(x,y) = \sum_{n=1}^{\infty} b_n(y) \sin(n\pi x)$$

$$1 = \sum_{n=1}^{\infty} \frac{2[1-(-1)^n]}{n\pi} \sin(n\pi x)$$

$$b_n''(y) - n^2\pi^2 b_n(y) = C_n \qquad b_n(0) = b_n(1) = 0$$

$$b_n = \frac{C_n}{n^2\pi^2} \left[\cosh(n\pi y) - 1 + \frac{\sinh(n\pi y)(1-\cosh n\pi)}{\sinh(n\pi)} \right.$$

$$u(x,y) = \sum_{n=1}^{\infty} b_n(y) \sin(\pi n x)$$

(b) Where by the symmetry, $u(y,x)$ is also a solution.

8. For $u_t = u_{xx} - u$

$$u(0,t) = 1, \quad u(1,t) = e^{-t}$$

$$u(x,0) = 1 \quad x \in (0,1)$$

(a) The steady state solution. $\lim_{t\to\infty} u(x,t) = U(x)$

$$U''(x) = U(x) \qquad U(0) = 1 \qquad U(1) = \lim_{n\to\infty} e^{-t} = 0$$

$$V = \cosh(x) - \coth(1)\sinh(x)$$

(b) $u(x,t) = 1 + (e^{-t} - 1)x + v(x,t)$

$$-xe^{-t} + v_t = v_{xx} - 1 - (e^{-t} - 1)x - v$$

$$v_t = v_{xx} - 1 + x - v$$

$v(0,t) = 0$

$v(1,t) = 0$

$v(x,0) = 0$

$v = X \cdot T$

$$X \cdot T' = X'' T \qquad - XT$$

$$1 + \frac{T'}{T} = \frac{X''}{X} = -\lambda$$

$$X_n = \sin(n\pi x)$$

$$v(x,t) = \sum_{n=1}^{\infty} b_n(t) \sin(n\pi x)$$

$$x - 1 = \sum_{n=1}^{\infty} E_n \sin(n\pi x)$$

Omit.

(c)

$$U(x) = \lim_{t \to \infty} = 1 - x - \sum \frac{2\sin(n\pi x)}{n\pi(1 + n^2\pi^2)}$$

$$\sum \frac{\sin(n\pi x)}{n\pi(n^2\pi^2 + 1)} = \frac{1}{2}\left[1 - x - \cosh(x) + \coth(1)\sinh(x)\right]$$

PS 7

7. $\mathcal{L}\{f(t)\sin(wt)\} = \frac{1}{2i}\{F(s-iw) - F(s+iw)\}$

$\mathcal{L}\{f(t)\cos(wt)\} = \frac{1}{2}(F(s-iw) + F(s+iw))$

12. $y'(t) = a\,y(t-T)$

$y(t) = 1 \qquad -T \le t \le 0$

(a)

$$\mathcal{L}\{y(t-T)\} = \int_0^\infty y(t-T)\,e^{-st}\,dt = e^{-sT}Y(s) + \frac{1-e^{-sT}}{s}$$

$$Y(s) = \frac{1+\alpha(1-e^{-sT})}{s(s-a\,e^{-sT})}$$

(b)

$$Y(s) = \frac{1+\alpha}{s^2}\cdot\sum_{n=0}^{\infty}\frac{a^n e^{-nsT}}{s^n} - \frac{\alpha\,e^{-sT}}{s^2}\sum_{n=0}^{\infty}\frac{\alpha^n e^{-nsT}}{s^n}$$

$$y(t) = (1+a)\sum_{n=0}^{\infty}\frac{\alpha^n (t-nT)^{n+1} H(t-nT)}{(n+1)!} - \alpha\sum_{n=0}^{\infty}\frac{\alpha^n[t-(n+1)T]^{n+1} H[t-(n+1)T]}{(n+1)!}$$

PS 8.

12

$$\int_{-\infty}^{\infty}|f(x)|^2\,dx = \frac{1}{2\pi}\int_{-\infty}^{\infty}|\hat{f}(\tau)|^2\,d\tau$$

1. $r(r-1) - \frac{1}{2}r = 0$

$r = 0$ or $r = \frac{3}{2}$

$$y = \sum_{n=0}^{\infty} a_n x^{n+r}$$

$$\sum_{n=0}^{\infty} x^{n+r} \left[2(n+r+1)(n+r)\, a_{n+1} - (n+r+1)\, a_{n+1} + (n+r) a_n + a_{n-1} \right] = 0$$

$$2(r+1) r\, a_1 - (r+1) a_1 + r a_0 = 0$$

$$a_1 = \frac{-r}{2r^2 + r - 1} a_0$$

When $r = 0$

$$2(n+1) n\, a_{n+1} - (n+1) a_{n+1} + n a_n + a_{n-1} = 0$$

$$\begin{cases} a_1 = 0 \\ a_0 = 1 \end{cases}$$

$$4a_2 - 2a_2 + a_1 + a_0 = 0$$

$$a_2 = -\frac{1}{2}$$

$$12 a_3 - 3a_3 + 2a_2 + a_1 = 0$$

$$a_3 = \frac{1}{9}$$

$$y_1 = 1 - \frac{1}{2}x^2 + \frac{1}{9}x^3 + \cdots$$

When $r = \frac{3}{2}$

$$(2n+5)\left(n + \frac{3}{2}\right) a_{n+1} - \left(n + \frac{5}{2}\right) a_{n+1} + \left(n + \frac{3}{2}\right) a_n + a_{n-1} = 0$$

$$a_1 = \frac{-r}{(2r-1)(r+1)} a_0 = \frac{-\frac{3}{2}}{2 \cdot \frac{5}{2}} a_0 = -\frac{3}{10} a_0$$

$$\begin{cases} a_1 = -\frac{3}{10} \\ a_0 = 1 \end{cases}$$

$$7 \cdot \frac{5}{2} a_2 - \frac{7}{2} a_2 + \frac{5}{2} a_1 + a_0 = 0$$

$$a_2 = -\frac{1}{56}$$

$$y_2 = 1 - \frac{3}{10}x - \frac{1}{56}x^2 + \cdots$$

2. (a) $\mathcal{L}\{t y''(t)\} + \mathcal{L}\{t y'(t)\} + \hat{y}(s) = 0$

$$-\frac{d}{ds}\left[s^2\hat{y}(s) - s y(0) - y'(0)\right] - \frac{d}{ds}\left[s\hat{y}(s) - y(0)\right] + \hat{y}(s) = 0$$

$$-2\cancel{s}\,Y(s) - s^2 Y'(s) - \cancel{Y(s)} - \cancel{s}\,Y'(s) + \cancel{Y(s)} = 0$$

$$(1+s)Y'(s) = -2Y(s)$$

$$\frac{Y'(s)}{Y(s)} = -\frac{2}{1+s}$$

$$\ln Y(s) = -2\ln(1+s) + C$$

$$Y(s) = \frac{C}{(1+s)^2}$$

$$y(s) = \mathcal{L}^{-1}\{Y(s)\} = C t e^{-t}$$

$$y'(s) = C(1-t)e^{-t}$$

$$y'(0) = C = 2$$

$$y(s) = 2t e^{-t}$$

(b) $$\mathcal{F}\{xe^{-|x|}\} = i\frac{d}{dk}\mathcal{F}\{e^{-|x|}\}$$
$$= i\frac{d}{dk}\left\{\frac{2}{1+k^2}\right\}$$
$$= -\frac{4ki}{(1+k^2)^2}$$

$$xe^{-|x|} = \frac{1}{2\pi}\int_{-\infty}^{\infty} -\frac{4ki}{(1+k^2)^2}\cdot e^{ikx}\,dk$$
$$= \frac{2}{\pi}\int_{-\infty}^{\infty} -\frac{ki\,(\cos kx + i\sin kx)}{(1+k^2)^2}\,dk$$
$$= \frac{4}{\pi}\int_{0}^{+\infty}\frac{k\sin kx}{(1+k^2)^2}\,dk$$
$$I = \frac{\pi}{4}e^{-1}$$

3. (a) $$\psi(0,t) = \cos t$$
$$\psi(1,t) = 0$$
$$\psi(x,0) = 1-x$$
$$\psi(x,t) = (1-x)\cos t$$
$$\frac{\partial v}{\partial t} - (1-x)\sin t = \frac{\partial^2 v}{\partial x^2} + 0$$

(b)

$$X\cdot T' = X''(x)\cdot T(t)$$
$$\frac{T'}{T} = \frac{X''}{X} = -\lambda$$

consider eigenfunction problem of $X'' = -\lambda X$ with $X(0)=0$
$X(1)=0$

$$X = \sin(n\pi x)$$

$$v(x,t) = \sum_{n=1}^{\infty} a_n(t)\sin(n\pi x)$$

$$\sum_{n=1}^{\infty} a_n'(t)\sin(n\pi x) - n^2\pi^2 a_n(t)\sin(n\pi x) = (1-x)\sin t = \sum_{n=1}^{\infty} b_n(t)\sin(n\pi x)$$

$$b_n(t) = \int_0^1 (1-x)\sin t \cdot \sin(n\pi x)\,dx$$

$$= \sin t \frac{1}{n\pi}\left[(x-1)\cos(n\pi x)\Big|_0^1 - \int_0^1 \cos(n\pi x)\,dx\right] = \frac{2\sin t}{n\pi}$$

$$a_n'(t) - n^2\pi^2 a_n(t) = \frac{2}{n\pi}\sin t \qquad a_n(0)=0$$

$$b_n(t) = \frac{\frac{2}{n\pi}\sin t\ \exp(-n^2\pi^2 t) + d_n\left[n^2\pi^2\sin(t) - \cos t\right]}{n^4\pi^4+1}$$

4 (a) $$f(\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos(n\theta) + b_n\sin(n\theta)$$

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(\theta)\,d\theta$$

$$= \frac{1}{\pi}\left(\frac{\pi}{2} - \pi\right) = -\frac{1}{2}$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(\theta)\cos(n\theta)\,d\theta = \frac{1}{\pi}\left(\frac{1}{n}\sin n\theta\Big|_{\frac{\pi}{2}}^{\pi} - \frac{1}{n}\sin n\theta\Big|_{\pi}^{2\pi}\right)$$

$$= -\frac{1}{n\pi}\sin\frac{n\pi}{2}$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(\theta)\sin(n\theta)\,d\theta = \frac{1}{\pi}\left(-\frac{1}{n}\cos n\theta\Big|_{\frac{\pi}{2}}^{\pi} + \frac{1}{n}\cos n\theta\Big|_{\pi}^{2\pi}\right)$$

$$= \frac{1}{n\pi}\left(1 - 2\cos n\pi + \cos\frac{n\pi}{2}\right)$$

(b)

$$\frac{1}{2} = -\frac{1}{4} + \sum_{n=1}^{\infty} \cancel{-\frac{1}{n\pi}\sin\left(\frac{n\pi}{2}\right)\cos\left(\frac{n\pi}{2}\right)} + \frac{1}{n\pi}\left[\cancel{\cos\left(\frac{n\pi}{2}\right)} + 1 - 2(-1)^n\right].$$

$$\sin\left(\frac{n\pi}{2}\right)$$

$$\frac{3}{4} = \sum_{n=1}^{\infty} \frac{1}{n\pi}\left[1 - 2(-1)^n\right]\sin\left(\frac{n\pi}{2}\right)$$

$$RHS = \frac{3\pi}{4}$$

5. (a) use superposition. since linear.

(b) $B = U'(r)$

$$B' + \frac{B}{r} = 0$$

$$B = r$$

$$B' + \frac{B}{r} = 4$$

$$[rB]' = 4r$$

$$rB = 2r^2 + C$$

$$B = 2r + \frac{C}{r}$$

$$U = r^2 + c_1 \ln r + C_2$$

$C_1 = 0$ since boundness

$$C_2 = -4$$

$$U = r^2 - 4$$

(b) $V(r,\theta) = R(r)\cdot\Theta(\theta)$

$$R''\theta + \frac{1}{r}R'\theta + \frac{1}{r^2}R\theta'' = 0$$

$$\frac{\Theta''}{\theta'} = \frac{R'' + \frac{1}{r}R'}{-\frac{1}{r^2}R} = -\lambda$$

$$\Theta_n(\theta) = \sin(n\theta) \text{ or } \cos(n\theta) \qquad \lambda = n^2$$

$$\Theta_0(\theta) = A_0$$

$$R'' + \frac{1}{r}R' - \frac{\lambda}{r^2}R = 0$$

$$R_n = A_n r^n + B_n r^{-n}$$

$$R_0 = A_0 + B_0 \ln r$$